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A Cross-Layer SDN Orchestration Architecture for Stateful Edge Services: Design and Experimental Characterization

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Master's in Telecommunications and Computer Engineering

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Month, Year

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TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

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Acknowledgments

Write here your acknowledgements and, if applicable, any grants or sources of funding.

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Resumo

Palavras-Chave:

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Abstract

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Keywords: *Keywords; separated; by; semicolons*

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Contents

Acknowledgments	iii
Resumo	v
Abstract	vii
List of Figures	xi
List of Tables	xiii
List of Acronyms	xv
Chapter 1. Introduction	1
1.1. Context, Motivation, and Problem Statement	1
1.2. Objectives	1
1.3. Research Questions	1
1.4. Research Methodology	1
1.5. Main Contributions of this Dissertation	1
1.6. Dissertation Structure	1
Chapter 2. Background and Related Work	3
2.1. Review Methodology	3
2.2. The Unexamined Default: Three-Layer Separation	3
2.3. Auto Scaling	3
2.4. Load Balancing on SDN	3
2.5. Monitoring & Telemetry	3
2.6. Resource Orchestration on SDN	3
2.7. Summary and Research Gaps	3
Chapter 3. Architecture and Design of Proposed Solution for Edge Services	5
3.1. Design Requirements	5
3.2. Architecture	5
3.3. Distributed Elastic Allocation of Resources	5
3.3.1. Services	5
3.3.2. Data	5
3.4. Monitoring and Decision Engine	5
3.5. Control Workflow	5
Chapter 4. Implementation of Proposed Solution	7

4.1. Experimental Infrastructure	7
4.2. Distributed Elastic Allocation of Resources	7
4.2.1. Services	7
4.2.2. Data	7
4.3. Monitoring and Decision Engine	7
4.4. Control Workflow	7
4.5. Implementation Validation	7
Chapter 5. Evaluation Methodology	9
5.1. Experimental Objectives	9
5.2. Evaluation Scenarios	9
5.3. Performance Metrics	9
5.4. Experimental Procedure with Results Statistical Analysis	9
Chapter 6. Experimental Results	11
6.1. Trigger Quality — Overload Detection (RQ3)	11
6.2. Telemetry Freshness — Delivery Cadence (RQ1)	11
6.3. Backend Selection — Routing Awareness (RQ2)	11
6.4. Network Performance	11
6.5. Scalability Analysis	11
6.6. Discussion — The Compound Coordination Gap	11
Chapter 7. Conclusions and Future Work	13
7.1. Conclusions	13
7.2. Research Contributions	13
7.3. Limitations	13
7.4. Future Research Directions	13

List of Figures

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List of Tables

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List of Acronyms

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CHAPTER 1

Introduction

Orchestrating edge web services means dealing with coordination delays between routing, auto-scaling and monitoring components. This has been shown to negatively affect overhead on overall infrastructure and QoS due to the unaccounted effects each module has on each other. In this chapter the context, motivation and problem statement are explained in the 1.1 with the objectives presented in 1.2. The research questions that emerged are available on 1.3. The methodology used to develop the thesis is detailed in 1.4, the main contributions in 1.5 and finally the outlining of the rest of the 7 chapters of the thesis are presented in 1.6.

1.1. Context, Motivation, and Problem Statement

1.2. Objectives

1.3. Research Questions

1.4. Research Methodology

1.5. Main Contributions of this Dissertation

1.6. Dissertation Structure

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CHAPTER 2

Background and Related Work

- 2.1. Review Methodology
- 2.2. The Unexamined Default: Three-Layer Separation
- 2.3. Auto Scaling
- 2.4. Load Balancing on SDN
- 2.5. Monitoring & Telemetry
- 2.6. Resource Orchestration on SDN
- 2.7. Summary and Research Gaps

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CHAPTER 3

Architecture and Design of Proposed Solution for Edge Services

This chapter presents the design of the proposed cross layer sdn orchestration architecture for edge services. It stems from the research gaps explored in the 2 and brings to the fold the need for an architecture that allows for each mentioned component to be independently managed and tested.

3.1. Design Requirements

3.2. Architecture

3.3. Distributed Elastic Allocation of Resources

3.3.1. Services

3.3.2. Data

3.4. Monitoring and Decision Engine

3.5. Control Workflow

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CHAPTER 4

Implementation of Proposed Solution

4.1. Experimental Infrastructure

4.2. Distributed Elastic Allocation of Resources

4.2.1. Services

4.2.2. Data

4.3. Monitoring and Decision Engine

4.4. Control Workflow

4.5. Implementation Validation

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CHAPTER 5

Evaluation Methodology

5.1. Experimental Objectives

5.2. Evaluation Scenarios

5.3. Performance Metrics

5.4. Experimental Procedure with Results Statistical Analysis

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CHAPTER 6

Experimental Results

- 6.1. Trigger Quality — Overload Detection (RQ3)
- 6.2. Telemetry Freshness — Delivery Cadence (RQ1)
- 6.3. Backend Selection — Routing Awareness (RQ2)
- 6.4. Network Performance
- 6.5. Scalability Analysis
- 6.6. Discussion — The Compound Coordination Gap

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CHAPTER 7

Conclusions and Future Work

7.1. Conclusions

7.2. Research Contributions

7.3. Limitations

7.4. Future Research Directions